Type MS · When is the particle at rest (or at a given velocity)?

Answer 1 of 16 · Question 5 part a · 3 marks

2022 JAN · Paper 02 · extract 1 of 3 · find part a in this mark scheme extract

5(a)	$3t^2-23t+30=0\Rightarrow(3t-5)(t-6)=0\Rightarrow t=6,\quad \frac{5}{3}$	M1A1A1 [3]
(b)	$a=\frac{\mathrm{d}v}{\mathrm{d}t}=6t-23$ $(6t-23>0)\Rightarrow t>\frac{23}{6}$	M1 A1 [2]
(c)	$s=\int 3t^2-23t+30\ \mathrm{d}t=\frac{3t^3}{3}-\frac{23t^2}{2}+30t+c$ $26=\frac{3(8)^3}{3}-\frac{23(8)^2}{2}+30(8)+c\Rightarrow c='10'$ $d=0+0+0+'10'\Rightarrow d=10$	M1A1 M1 A1 [4]
	Total	**9 marks**

Type MS · When is the particle at rest (or at a given velocity)?

Answer 1 of 16 · Question 5 part a · 3 marks

2022 JAN · Paper 02 · extract 2 of 3 · find part a in this mark scheme extract

Part	Mark	Notes
5 (a)	M1	For setting the given expression for $v = 0$ and attempting to solve the 3TQ See general guidance for the definition of an attempt $3t^2 - 23t + 30 = 0 \Rightarrow (3t-5)(t-6) = 0 \Rightarrow t = ..., ...$
	A1	For either $t = 6$ or $t = \frac{5}{3}$
	A1	For both $t = 6$ and $t = \frac{5}{3}$
(b)	M1	For differentiating the given expression for v which must be correct for this mark. $a = \frac{dv}{dt} = 6t - 23$
	A1	$(6t - 23 > 0) \Rightarrow t > \frac{23}{6}$ Accept equivalent exact values including $t > 3.8\dot{3}$
(c)	M1	For attempting to integrate the given expression for v See General Guidance for the definition of an attempt with no terms differentiated. $+c$ is not required for the award of this mark. $s = \int 3t^2 - 23t + 30 \, dt = \frac{3t^3}{3} - \frac{23t^2}{2} + 30t + c$
	A1	For the correct integrated expression, which must include a constant term e.g.+ c $s = \frac{3t^3}{3} - \frac{23t^2}{2} + 30t + c$ Simplification is not required for this mark
	M1	For substituting the given values of $t = 8$ when $s = 26$ into their **integrated** expression to find c $26 = \frac{3(8)^3}{3} - \frac{23(8)^2}{2} + 30(8) + c \Rightarrow c = '10'$ $[d = 0 + 0 + 0 + '10' \Rightarrow d = ...]$
	A1	For the correct value of $d = 10$

Type MS · When is the particle at rest (or at a given velocity)?

Answer 1 of 16 · Question 5 part a · 3 marks

2022 JAN · Paper 02 · extract 3 of 3 · find part a in this mark scheme extract

Question	Scheme	Mark

Type MS · When is the particle at rest (or at a given velocity)?

Answer 2 of 16 · Question 3 part b · 3 marks

2023 NOV · Paper 02 · extract 1 of 3 · find part b in this mark scheme extract

Question	Scheme	Marks
3(a)(i)	$v = 6t^2 - 16t + c$	M1
	$t = 0, \quad v = 12$	A1
	$v = \frac{6t^2}{2} - 16t + 12 = \left[3t^2 - 16t + 12\right]$	M1
(ii)	$s = \frac{3t^3}{3} - \frac{16t^2}{2} + 12t + k$	A1
	$\left[t = 0, \quad s = 0\right] \Rightarrow k = 0$	[4]
	$s = t^3 - 8t^2 + 12t$	
(b)	At the origin, $s = 0$	
	$t^3 - 4t^2 + 12t = 0$	M1
	$\Rightarrow t(t-2)(t-6) = 0$	dM1
	$\Rightarrow t = 2, 6, (0)$	A1
	P first returns to the origin when $t = 2$ seconds	[3]
		Total 7 marks

Type MS · When is the particle at rest (or at a given velocity)?

Answer 2 of 16 · Question 3 part b · 3 marks

2023 NOV · Paper 02 · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Notes
(a)(i)	M1	For integrating the given expression. ($+c$) not required for this mark See General Guidance for the definition of an attempt. No term to be differentiated.
	A1	Finds the value of c **and** the correct expression for v
(ii)	M1	For an acceptable attempt to integrate **their** v. See General Guidance for the definition of an attempt. ($+k$) not required for this mark. No term to be differentiated.
	A1	For the correct expression for the distance, which must have come from establishing **explicitly** that $k = 0$ from the integration. Without $+ k$ in their integration this mark is automatically lost.
(b)	M1	For setting their expression [which must be a cubic] for $s = 0$
	dM1	For solving the cubic Minimally acceptable attempt is $\Rightarrow t(t \pm A)(t \pm B) = 0$ where $\lvert AB \rvert = 12$ or also allow $(t \pm A)(t \pm B) = 0$ where $\lvert AB \rvert = 12$ as candidates are told that one value of $t = 0$ Any method to solve the quadratic is acceptable. Formula or completing the square. This is a dependent mark. <u>Use of a calculator</u> If there is no working seen and either their values of t are incorrect or their s is incorrect, only award this mark if a correct method is used/seen to solve the equation.
	A1	For the correct value of $t = 2$ explicitly stated.

Type MS · When is the particle at rest (or at a given velocity)?

Answer 2 of 16 · Question 3 part b · 3 marks

2023 NOV · Paper 02 · extract 3 of 3 · find part b in this mark scheme extract

Question	Scheme	Marks

Type MS · When is the particle at rest (or at a given velocity)?

Answer 3 of 16 · Question 4 part b · 2 marks

2019 JUN · Paper 02 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

4 (a)	$v = \int (6t-12)\ \mathrm{d}t = 3t^2 - 12t\ (+c)$ When $t = 0$, $v = 0$ so $c = 0$ $v = 3t^2 - 12t = 3\times 2^2 - 12\times 2 = -12\ (\text{m/s})$	M1A1 A1cso [3]
(b)	$v = 0,\ 3t^2 - 12t = 0 \Rightarrow 3t(t-4) = 0 \Rightarrow T = 4$	M1A1 [2]
(c)	$s = \int (3t^2 - 12t)\ \mathrm{d}t = t^3 - 6t^2\ (+k)$ $d = \left\lvert \left[t^3 - 6t^2\right]_0^4 \right\rvert + \left[t^3 - 6t^2\right]_4^8 = 32 + 160 = 192\ (\text{m})$	M1 M1A1 [3]
Alternative		
(c)	$s = \int (3t^2 - 12t)\ \mathrm{d}t = t^3 - 6t^2\ (+k)$ At time $t = 0$, distance = 0, $k = 0$ $t = 4 \quad s = 4^3 - 6\times 4^2 = 64 - 96 = -32$ $t = 8 \quad s = 8^3 - 6\times 8^2 = 512 - 384 = 128$ Total distance travelled = 32 +32 +128 = 192 (m)	M1 M1 A1
		Total 8 marks

(a)	M1	Attempt to integrate $6t - 12$, must see power of t increased by one in either term and no decrease in powers. Do not accept $v = t(6t-12)$.
	A1	Correct result of integration. Constant is not required.
	A1cso	Correct velocity. Negative sign is required. Must show or state constant = 0.
(b)	M1	$3t^2 - 12t = 0$ and attempt to solve. ft their integration if M1 awarded in (a).
	A1	$T = 4$ Allow $t = 4$. Do not allow if $T = 0$ is also given as an option.
(c)	M1	Attempt to integrate their velocity providing it includes at least one term in t^2 or higher powers. Must see power of t increased by one in at least one term and no decrease in powers. Do not accept $s = t\times$ their v.
	M1	Attempt to use limits of 0 to 4 and 4 to 8 and then add the magnitudes of the two values. ft their $T = 4$ if M1 awarded in (b).
	A1	192
Alternative		
(c)	M1	Attempt to integrate their velocity providing it includes at least one term in t^2 or higher powers. Must see power of t increased by one in at least one term and no decrease in powers. Do not accept $v = t(6t-12)$.
	M1	Substitute and attempt to find distance when $t = 4$ and when $t = 8$. Working to confirm $k = 0$ must be shown. ft their $T = 4$ if M1 awarded in (b).
	A1	192

Type MS · When is the particle at rest (or at a given velocity)?

Answer 3 of 16 · Question 4 part b · 2 marks

2019 JUN · Paper 02 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · When is the particle at rest (or at a given velocity)?

Answer 4 of 16 · Question 1 part whole · 4 marks

2020 NOV · Paper 02 · extract 1 of 9 · find part whole in this mark scheme extract

1	$(v=)8+2t-t^2$ $8+2t-t^2=(2+t)(4-t)=0\Rightarrow t=4$ Distance $=3+8\times4+4^2-\frac{1}{3}4^3=29\frac{2}{3}$ m (accept 29.7 or better or a recurring decimal)	B1 M1A1 A1 (4) [4]

B1	Correct differentiation
M1	Equate their differentiated expression (min 2 correct terms) to 0 (= 0 may be implied by their solution) and attempt to solve the 3 TQ by any valid method. Must reach t = ... Calculator solution: Allow M1A1 if their equation and its roots are correct , otherwise M0A0
A1	Correct value of t (Ignore t = -2 if shown)
A1	Correct distance, exact or min 3 s f Award A0 if value when t = -2 is also offered (and not excluded) If there is an error in the solution of their equation but t = 4 is used to obtain the correct answer this mark cannot be awarded.

Type MS · When is the particle at rest (or at a given velocity)?

Answer 4 of 16 · Question 1 part whole · 4 marks

2020 NOV · Paper 02 · extract 2 of 9 · find part whole in this mark scheme extract

Question Number	Scheme	Marks

Type MS · When is the particle at rest (or at a given velocity)?

Answer 4 of 16 · Question 1 part whole · 4 marks

2020 NOV · Paper 02 · extract 3 of 9 · find part whole in this mark scheme extract

ALT	For second and third marks: $\frac{5}{ND} = 10 \Rightarrow ND = \frac{1}{2}$ $OD = \frac{1}{2}\ln 5 - \frac{1}{2}$	 M1 A1	 [15]

(a)	
M1	Equate the 2 curve equations. No need to simplify
M1	Factorise their equation
A1	Obtain the one possible value for x (other root need not be seen; if seen it must be rejected) **Must be exact**
A1	Obtain the corresponding value for y. **Must be exact.** Need not be shown in coordinate brackets. Use of $e^{2x} = 5$ leads to $y = 5$ without use of a value of x, so M1M1A0A1 can be scored. There must only be one correct y shown. Accept $y = e^{\ln 5}$
(b)	
M1	Differentiate the equation of C_1 $5e^{-2x} \rightarrow ke^{-2x}$ where $k = \pm 5$ or ± 10 and no integration seen
A1ft	Grad at A = -2 follow through their x coordinate
dM1	Obtain the equation of the tangent at A using their gradient and their coordinates of A. Can be in any form but if $y = mx + c$ is used a value for c must be found. Gradient of the tangent must be numerical.
A1	Correct equation in any form
A1cso	Correct x coordinate of B obtained from correct working.
ALT	For last 3 marks
dM1	Use their y coordinate of A and their (numerical) gradient of the tangent to find the length NB (where N is the foot of the perpendicular from A to the x-axis)
A1	Correct length of NB
A1cso	Add the x coordinate of A to obtain the x coordinate of B
(c)	
B1ft	Correct gradient of tangent to C_2 at A follow through their x coordinate
M1	Obtain an equation for the tangent using their gradient and their coordinates of A Gradient of the tangent must be numerical.
A1	Correct x coordinate of D (exact or minimum 3 sf)
M1	Use a correct formula for the area of a triangle with their y coordinate of A, their x coordinate of D and the **given** x coordinate of B
A1	Correct, unsimplified area Allow use of correct but non-exact coordinates
A1	Correct area Accept only $7\frac{1}{2}, \frac{15}{2}$ or 7.5
	Heron's formula: Nos which may be seen: $AB = \frac{5\sqrt{5}}{2}, AD = \frac{\sqrt{101}}{2}, BD = 3, s = \frac{1}{2}(a+b+c) = 6.8$
ALT	For second and third marks:
M1	Use their y coordinate of A and their gradient of the tangent to find the length ND
A1	Use the x coordinate of A to obtain the x coordinate of D

Type MS · When is the particle at rest (or at a given velocity)?

Answer 4 of 16 · Question 1 part whole · 4 marks

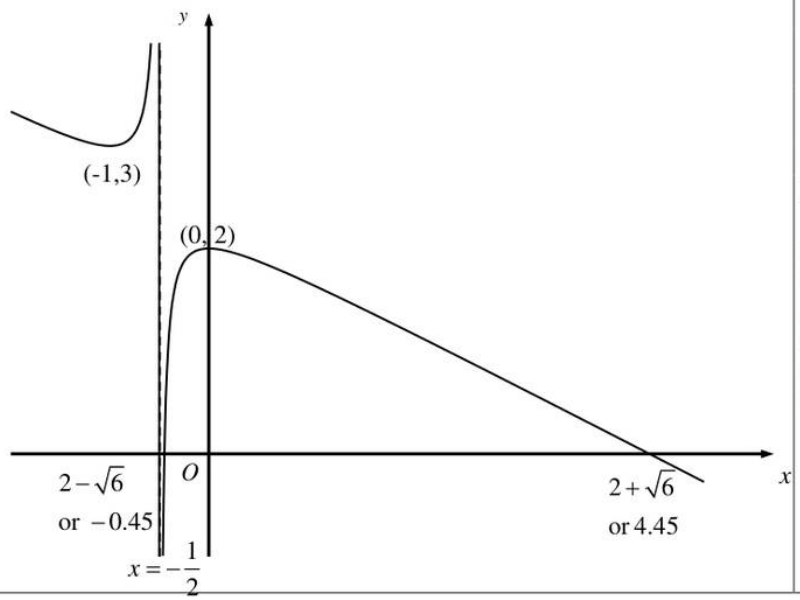
2020 NOV · Paper 02 · extract 4 of 9 · find part whole in this mark scheme extract

ALT	Area by "determinant" method:
M1	Eg Area $= \frac{1}{2}\begin{vmatrix} \frac{1}{2}\ln 5 & \frac{1}{2}(5+\ln 5) & \frac{1}{2}(\ln 5-1) & \frac{1}{2}\ln 5 \\ 5 & 0 & 0 & 5 \end{vmatrix}$ y coordinates of B and D must be 0 Must include the ½ and have 4 sets of coordinates with first and last the same.
A1	$= \frac{1}{2}\left(\frac{1}{2}(\ln 5-1)\times 5 - \frac{1}{2}(\ln 5+1)\right)$ Allow use of correct but non-exact coordinates
A1	Correct area Accept only $7\frac{1}{2}, \frac{15}{2}$ or 7.5 **Must be positive**

Type MS · When is the particle at rest (or at a given velocity)?

Answer 4 of 16 · Question 1 part whole · 4 marks

2020 NOV · Paper 02 · extract 5 of 9 · find part whole in this mark scheme extract

Question Number	Scheme	Marks
9(a)	$y=\dfrac{2+4x-x^2}{2x+1} \Rightarrow x^2-4x-2+2yx+y\ (=0)$	
	$x^2+(2y-4)x+(y-2)=0$	M1A1A1 (3)
(b)	$(2y-4)^2 \geq 4(y-2)$	M1
	$4y^2-16y+16=4y-8 \Rightarrow 4y^2-20y+24=0$	M1A1
	$y \leq 2$ or $y \geq 3$ *	A1cso (4)
(c)	$y=\dfrac{2+4x-x^2}{2x+1}$	
	$\dfrac{dy}{dx}=\dfrac{(4-2x)(2x+1)-2(2+4x-x^2)}{(2x+1)^2}$ **See notes for product rule method**	M1A1A1
	$\dfrac{dy}{dx}=0 \Rightarrow (4-2x)(2x+1)-2(2+4x-x^2)=0$	
	$2x(x+1)=0 \Rightarrow x=0,\ -1$	M1 A1
	stationary points are $(0,2)$ $(-1,3)$	A1 (6)
ALT	$x^2+(2y-4)x+(y-2)=0 \Rightarrow 2x+(2y-4)+2\dfrac{dy}{dx}x+\dfrac{dy}{dx}=0$	M1A1A1
	$\dfrac{dy}{dx}=0 \Rightarrow x+y=2$	M1
	(using (b)) stationary points are $(0,2)$ $(-1,3)$	A1A1
(d)		B1 curve (i) M1A1 (M1 finding coords, A1 correct (oe or min 2dp) and on diagram (ii) B1 (iii) B1ft (5) [18]

Type MS · When is the particle at rest (or at a given velocity)?

Answer 4 of 16 · Question 1 part whole · 4 marks

2020 NOV · Paper 02 · extract 6 of 9 · find part whole in this mark scheme extract

(a)M1	Re-write the equation of C without fractions and rearrange to the required form.
A1	Correct value for a and either b or c. These values need not be stated explicitly.
A1	Fully correct equation. Values of a, b and c need not be stated explicitly. Condone missing brackets round "$y-2$". Award this mark when the equation is reached – isw any listing of values with incorrect signs.
(b)	
M1	Use "$b^2 \geqslant 4ac$" for their equation
M1	Re-arrange their inequality to a 3TQ in y. Allow an equation here.
A1	Correct 3TQ, as shown or any equivalent
A1cso	Deduce the CVs (need not be shown explicitly) and state the (given) inequalities. There must be no errors in the working but the equation, if correct, can be solved easily so no working need be shown. Condone use of "and" instead of "or".
(c)	
M1	Differentiate the equation of C using the quotient rule. The denominator must be correct and the numerator must consist of the difference of 2 terms of the type shown. The product rule can be used.
A1	Either numerator term correct
A1	Fully correct numerator
ALT:	**Product Rule** $y=\left(2+4x-x^2\right)\left(2x+1\right)^{-1}$
M1A1 A1	$\frac{\mathrm{d}y}{\mathrm{d}x}=\left(4-2x\right)\left(2x+1\right)^{-1}-2\left(2+4x-x^2\right)\left(2x+1\right)^{-2}$ M1: rewrite without denominator and attempt product rule. Difference of 2 terms of the form shown needed (Difference because of the negative power) A1 Either term correct A1 Second term correct
M1	Equate the numerator of their derivative to 0 and solve to x = ...(any valid method of solving a quadratic with 2 or 3 terms) If product rule used must multiply through by $\left(2x+1\right)^2$
A1	Both x values correct
A1	Both stationary points correct
ALT	
M1	Use implicit differentiation on the re-arranged equation
A1	Correct derivative of $\left(2y-4\right)x$ (inc use of product rule)
A1	Fully correct derivative
M1	Set $\frac{\mathrm{d}y}{\mathrm{d}x}=0$ and use the result from (b) to obtain solutions
A1	One correct stationary point
A1	Both correct stationary points
(d)	
B1	Shape: Two parts, one above $y = 3$ and the other below $y = 2$
(i)M1	Attempt to find the x coordinates of the crossing points
A1	Correct coordinates shown on their sketch, 2 crossing points only. $y = 0$ need not be seen. There must be a curve through these points.
(ii)B1	The asymptote must be drawn and labelled (by its equation or by showing the x coordinate of the point where it crosses the x-axis). There must be at least one part of the curve which is asymptotic to the line $x=-\frac{1}{2}$. No part of the curve should touch/cross the asymptote or curve dramatically away from the line.
(iii)B1ft	Label the stationary points with their coords. Follow through provided the result is sensible.

Type MS · When is the particle at rest (or at a given velocity)?

Answer 4 of 16 · Question 1 part whole · 4 marks

2020 NOV · Paper 02 · extract 7 of 9 · find part whole in this mark scheme extract

Question Number	Scheme	Marks
10(a)	$\cos(A+B)+\cos(A-B)=\cos A\cos B-\sin A\sin B+\cos A\cos B+\sin A\sin B$	M1
	$=2\cos A\cos B$ *	A1 cso (2)
(b)	Let $A+B=P,\quad A-B=Q \Rightarrow A=\frac{1}{2}(P+Q),\quad B=\frac{1}{2}(P-Q)$	M1
	As $\cos(A+B)+\cos(A-B)=2\cos A\cos B$	
	$\cos P+\cos Q=2\cos\frac{1}{2}(P+Q)\cos\frac{1}{2}(P-Q)$ *	M1A1cso (3)
ALT		
	Let $A=\frac{1}{2}(P+Q),\quad B=\frac{1}{2}(P-Q)\Rightarrow A+B=P,\ A-B=Q$	M1
	As $2\cos A\cos B=\cos(A+B)+\cos(A-B)$	
	$2\cos\frac{1}{2}(P+Q)\cos\frac{1}{2}(P-Q)=\cos P+\cos Q$ *	M1A1cso
(c)	$\cos 5\theta+\cos 7\theta=2\cos 6\theta\cos\theta=0$	M1
	$\cos 6\theta=0\Rightarrow 6\theta=\frac{\pi}{2},\frac{3\pi}{2},\frac{5\pi}{2}\Rightarrow\theta=\frac{\pi}{12},\frac{\pi}{4}\left(\text{or }\frac{3\pi}{12}\right),\frac{5\pi}{12}$	A1A1
	$\cos\theta=0\Rightarrow\theta=\frac{\pi}{2}$	A1 (4)
(d)	$\cos 8x+2\cos 6x+\cos 4x=(\cos 8x+\cos 6x)+(\cos 6x+\cos 4x)$	
	$=2\cos 7x\cos x+2\cos 5x\cos x$	M1
	$=2\cos x(\cos 7x+\cos 5x)=2\cos x\times 2(\cos 6x\cos x),\ =4\cos 6x\cos^2 x$ *	dM1,A1cso (3)
ALT 1	$\cos 8x+2\cos 6x+\cos 4x=(\cos 8x+\cos 4x)+2\cos 6x$	
	$=2\cos 6x\cos 2x+2\cos 6x$	M1
	$=2\cos 6x(\cos 2x+1)=2\cos 6x(2\cos^2 x-1+1)=4\cos 6x\cos^2 x$ *	dM1A1cso (3)
ALT 2	Working from right to left	
	$4\cos 6x\cos^2 x=4\cos 6x\times\frac{1}{2}(\cos 2x+1)=2\cos 6x\cos 2x+2\cos 6x$	M1
	$=\cos 8x+\cos 4x+2\cos 6x$ *	dM1A1cso (3)

Type MS · When is the particle at rest (or at a given velocity)?

Answer 4 of 16 · Question 1 part whole · 4 marks

2020 NOV · Paper 02 · extract 8 of 9 · find part whole in this mark scheme extract

(e)	$\int_0^{\frac{\pi}{3}} \cos 6x \cos^2 x \, dx = \frac{1}{4}\int_0^{\frac{\pi}{3}} (\cos 8x + 2\cos 6x + \cos 4x) \, dx$	M1	
	$= \frac{1}{4}\left[\frac{1}{8}\sin 8x + \frac{1}{3}\sin 6x + \frac{1}{4}\sin 4x\right]_0^{\frac{\pi}{3}}$	A1	
	$= \frac{1}{4}\left[\frac{1}{8}\sin\frac{8\pi}{3} + \frac{1}{3}\sin 2\pi + \frac{1}{4}\sin\frac{4\pi}{3} - 0\right] = -\frac{\sqrt{3}}{64}$ (-0.02706... scores M1A0)	dM1A1	(4)
			[16]
ALT	For the first M mark Award M1 only when the integrand has been changed (by a valid method) to a function which can be integrated $\int_0^{\frac{\pi}{3}} \cos 6x \cos^2 x \, dx = \int_0^{\frac{\pi}{3}} \cos 6x \times \frac{1}{2}(\cos 2x + 1) \, dx$		
	$= \frac{1}{2}\int_0^{\frac{\pi}{3}} (\cos 6x \times \cos 2x + \cos 6x) \, dx = \frac{1}{2}\int_0^{\frac{\pi}{3}} \frac{1}{2}(\cos 8x + \cos 4x + 2\cos 6x) \, dx$	M1	
	Rest as above.		

(a)	
M1	Use the standard formulae. This is a "show that" question so these formulae must be written out in full. Both sides of the result must be seen although the working may appear between them as seen in the scheme.
A1cso	Final **given** result. Do not award if the expansions shown are not in the correct order (this suggests incorrect signs in the formulae).
(b)	
M1	Use $A+B=P,\quad A-B=Q$ to obtain A and B in terms of P and Q
M1	Substitute in the result from (a) to obtain an identity in P and Q only
A1cso	Correct result reached with no errors seen
ALT	Working right to left: Notes similar to above
(c)	
M1	Use the result from (b) to show that $(2)\cos 6\theta \cos(\pm\theta) = 0$
A1	Obtain one correct solution of $\cos 6\theta = 0$. Allow if in decimal form but must be radians.
A1	Two further correct solutions **and no more within the range.** If any of the 3 solutions of $\cos 6\theta = 0$ is not exact do not award this mark.
A1	State the solution of $\cos(\pm\theta) = 0$. Must be exact unless this is penalised above.
NB	If answers are given in degrees deduct 2A marks from any that would otherwise have been given. (Answers in degrees but then changed to radians are acceptable – mark the radian answers.) Ignore extra answers outside the stated range – any within are incorrect.
(d)	
M1	Use the result from (a) or (b) once. Allow if one or both "2"s are missing
dM1	Use the result from (a) or (b) again. Must include both "2"s this time. Depends on previous M mark.
A1cso	Reach the **given** result with no errors seen

Type MS · When is the particle at rest (or at a given velocity)?

Answer 4 of 16 · Question 1 part whole · 4 marks

2020 NOV · Paper 02 · extract 9 of 9 · find part whole in this mark scheme extract

ALT 1	
M1	Use the result from (a) or (b). Allow if the "2" is missing.
dM1	Factorise and use the *correct* double angle formula on cos2x Depends on the previous M mark
A1cso	Reach the **given** result with no errors seen
ALT 2	
M1	Use the *correct* half angle formula on $\cos^2 x$
dM1A1	M1: Use the result from (a) or (b)
cso	A1: reach the **given** result with no errors seen
(e)M1	Obtain a function which can be integrated **either** by using the **given** result from (d) **OR**. deriving the same result. Allow if 1/4 is missing. (Integration by Parts – send to review)
A1	Correct integration (must have included the ¼) Limits not needed for these 2 marks.
dM1	Correct use of the given limits. All sines are 0 at the lower limit so these need not be shown
A1	Correct final answer which must be **exact and stated as a single fraction**.

Type MS · When is the particle at rest (or at a given velocity)?

Answer 5 of 16 · Question 5 part b · 2 marks

2024 JUN · Paper 01 · extract 1 of 2 · find part b in this mark scheme extract

5(a)	$\left(v=\int(3t-4)\,\mathrm{d}t=\right)\frac{3t^2}{2}-4t+c$ (When $t=0$, $v=0 \Rightarrow c=0$) $\left(v=\right)\frac{3\times4^2}{2}-4\times4=8$ [m/s]	M1 M1A1 [3]
(b)	$0=\frac{3t^2}{2}-4t \Rightarrow t\left(\frac{3t}{2}-4\right)=0 \Rightarrow \frac{3t}{2}=4 \Rightarrow t=\frac{8}{3}$	M1A1 [2]
(c)	$\left(x=\int\left(\frac{3t^2}{2}-4t\right)\mathrm{d}t=\right)\frac{3t^3}{2\times3}-\frac{4t^2}{2}(+c)=\left[\frac{t^3}{2}-2t^2+c\right]$ When $t=0$ P is at the point with coordinates $(-10, 0)$ $\Rightarrow c=-10$ $\Rightarrow \left(x=\right)\frac{3^3}{2}-2\times3^2-10=-\frac{29}{2}$ (m)	M1 M1 dM1A1 [4]
		Total 9 marks

Part	Mark	Notes
(a)	M1	For a minimally acceptable attempt (see general guidance) to integrate the given expression. **At least one term correctly integrated.** Terms do not need to be simplified to attain this mark. No power of t to decrease.
	M1	For substituting the value of $t=4$ into their changed expression (general rule of marking unless precluded, this method mark may be implied from a correct answer, unless from incorrect working).
	A1	For $(v=)8$
We do not need to see the calculation of $c=0$ Note: substitution of $t=4$ into the expression for a also gives an answer of 8 – please watch out for this and do not mark this correct.		
(b)	M1	For setting their changed expression = 0 and a **fully correct** attempt to solve, leading to a value for t. Note a correct vale of t can imply this mark. If the quadratic is not the correct quadratic, method must be shown.
	A1	For the value of $t=\frac{8}{3}$, ignore $t=0$ is given. Accept answers which round to 2.7 or clear indication of 2.6 recurring.
(c)	M1	For a **minimally acceptable attempt** (see general guidance) to integrate their expression for v, which must be a **minimum 2 term** expression. Terms don't need to be simplified at this point. c does not need to be present. No power of t to decrease.
	M1	For correct substitution, into their changed expression, of $t=0$ and $P=-10$ to find the value of c. -10, if seen at any point will usually imply this mark. The mark is for substitution of the correct values, to find c, if they rearrange incorrectly, the mark can still be awarded.
	dM1	For substituting the value of $t=3$ into their changed expression, with their c. Dependent on the previous method mark.
	A1	For the displacement of $-\frac{29}{2}$ If distance of $\frac{29}{2}$ is given A0, no isw here.

Type MS · When is the particle at rest (or at a given velocity)?

Answer 5 of 16 · Question 5 part b · 2 marks

2024 JUN · Paper 01 · extract 2 of 2 · find part b in this mark scheme extract

Question	Scheme	Marks

Type MS · When is the particle at rest (or at a given velocity)?

Answer 6 of 16 · Question 4 part b · 2 marks

2025 NOV · Paper 02 · extract 1 of 3 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	B1	For $\frac{4x-3}{2x+1}$ or $\frac{2(2x+1)}{2x+1}-\frac{5}{2x+1}$ or $\frac{4x+2-5}{2x+1}$ **Both** correct asymptote equations seen will imply this mark.
(i)	B1	For $x=-\frac{1}{2}$ Independent of method accuracy mark, **so ignore any incorrect attempt to form a single fraction**
(ii)	B1	For $y=2$ Independent of method accuracy mark, **so ignore any incorrect attempt to form a single fraction**
(b)	B1ft	For the coordinates of the y-intercept, follow through their a or b and c from any expression of the form $\frac{4x+a}{2x+1}$ $a\neq 0$ or $\frac{4x+b+c}{2x+1}$ $b,c\neq 0$ Condone eg just $y=-3$ or $y=a$ without $x=0$ shown.
	B1ft	For the coordinates of the x-intercept, follow through their a as above. ie ft $(x=)-\frac{a}{4}$ Condone $x=\frac{3}{4}$ or $x=-\frac{a}{4}$ without $y=0$ shown.
Note on labelling: Answers presented in the correct order without labelling – we would award all marks. If labelling not present, we can't assume the candidate's thinking.		
(c)	B1	For a negative reciprocal curve drawn anywhere in the grid – there **must** be two branches present, they **must not** cross any asymptotes drawn or implied or cross each other and **must not obviously** 'bend back' on themselves. Mark intention.
	B1ft	Two clearly marked asymptotes, ft their asymptotes listed from part (a). Labelled as equations or clearly passing through correctly labelled numbers on the axes. Must be one section of negative reciprocal curve, tending towards these asymptotes. This branch must not obviously cross or bend back from any asymptotes drawn or implied. Mark intention. Ignore any other branch/curve/line drawn if there is one branch fulfilling these conditions.
	B1ft	Single curve (or even line) passing through the correct intercepts. Ft their coordinates from part (b) marked clearly on the graph as coordinates or crossing points on the axes. Ignore any other branch/curve/line present.
	The answers to (a) and (b) cannot be retrieved from (c)	

Type MS · When is the particle at rest (or at a given velocity)?

Answer 6 of 16 · Question 4 part b · 2 marks

2025 NOV · Paper 02 · extract 2 of 3 · find part b in this mark scheme extract

Question	Scheme	Marks
4(a)	$\left(\frac{dv}{dt}=\right)8t-6=0\Rightarrow t=\left(\frac{6}{8}\right)$ oe	M1
	$v=4\times\left("\frac{6}{8}"\right)^2-6\times\left("\frac{6}{8}"\right)+5=\frac{11}{4}[\text{m/s}]$ oe	dM1A1 [3]
ALT	$4\left(t-\frac{3}{4}\right)^2+\frac{11}{4}$ oe	M1 M1
	$\frac{11}{4}[\text{m/s}]$ oe	A1 [3]
(b)	$18=8T-6\Rightarrow T=3$	M1A1 [2]
(c)	$\left[XY=\int_0^3\left(4t^2-6t+5\right)dt=\right]$	
	$\left[\left[\frac{4t^3}{3}-\frac{6t^2}{2}+5t\right]_0^3\right]$	M1A1
	$XY=\left(\frac{4\times3^3}{3}-\frac{6\times3^2}{2}+5\times3\right)-(0)=24\,(\text{m})$	M1A1 [4]
		Total 9 marks

Type MS · When is the particle at rest (or at a given velocity)?

Answer 6 of 16 · Question 4 part b · 2 marks

2025 NOV · Paper 02 · extract 3 of 3 · find part b in this mark scheme extract

Question	Scheme	Marks

Type MS · When is the particle at rest (or at a given velocity)?

Answer 7 of 16 · Question 5 part b · 2 marks

2023 JUN · Paper 02 (regional R) · extract 1 of 5 · find part b in this mark scheme extract

5(a)	$\frac{dv}{dt} = 4t - 19$ $\frac{dv}{dt} = 4\times 5 - 19 = 1\left(\text{m/s}^2\right)$	M1 A1 [2]
(b)	$2t^2 - 19t + 35 = 0 \Rightarrow (2t-5)(t-7) = 0 \Rightarrow t = \frac{5}{2}, 7$ $t_1 = \frac{5}{2} \quad t_2 = 7$	M1 A1 [2]
(c)	$D = \int_{\frac{5}{2}}^{7} \left(2t^2 - 19t + 35\right) dt = \left[\frac{2t^3}{3} - \frac{19t^2}{2} + 35t\right]_{\frac{5}{2}}^{7}$ $\left(\frac{2\times 7^3}{3} - \frac{19\times 7^2}{2} + 35\times 7\right) - \left(\frac{2\times 2.5^3}{3} - \frac{19\times 2.5^2}{2} + 35\times 2.5\right)$ $= \left[\frac{49}{6} - \frac{925}{24}\right] = [8.167 - 38.542] = \left[-\frac{243}{8} \text{ or } -30.375\right]$ $\Rightarrow D = \frac{243}{8}$ (m) oe e.g. 30.375	M1 M1 A1 [3]
		Total 7 marks

Type MS · When is the particle at rest (or at a given velocity)?

Answer 7 of 16 · Question 5 part b · 2 marks

2023 JUN · Paper 02 (regional R) · extract 2 of 5 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	M1	For differentiating the given v and substituting in $t = 5$ into the derivative. Both differentiation and substitution must be correct for this mark.
	A1	For the correct acceleration of 1 m/s^2. Units are not required.
(b)	M1	For attempting to solve the given 3TQ for the velocity. A correct method must be used, and they must reach **two** values of t for the award of this mark. See General Guidance for the definition of an attempt to solve a 3TQ. If there is no visible method seen, both values of t must be seen for the award of this mark.
	A1	For the two correct values of t They do not need to ne identified as t_1 or t_2 for this mark. Accept $\frac{5}{2}$ and 7 seen.
(c)	M1	For an attempt to integrate the given expression for v. At least one term must be correct and no terms are to be differentiated. If the value of 35 'disappears' it is M0. The question states 'use calculus' so integration must be seen.
	M1	For substituting the **two** values of t into their integrated expression. and subtracting the result (either way) of both substitutions. Allow for any changed expression from v A correct answer of $\pm\frac{243}{8}$ or ± 30.375 implies correct substitution into a correct integral. If the integration is incorrect, or the values are incorrect, full correct substitution must be seen for the award of this mark.
	A1	For the correct distance (which must be a positive value).

Type MS · When is the particle at rest (or at a given velocity)?

Answer 7 of 16 · Question 5 part b · 2 marks

2023 JUN · Paper 02 (regional R) · extract 3 of 5 · find part b in this mark scheme extract

Question	Scheme	Marks

Type MS · When is the particle at rest (or at a given velocity)?

Answer 7 of 16 · Question 5 part b · 2 marks

2023 JUN · Paper 02 (regional R) · extract 4 of 5 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	M1	For differentiating the given v and substituting in $t = 5$ into the derivative. Both differentiation and substitution must be correct for this mark.
	A1	For the correct acceleration of 1 m/s^2. Units are not required.
(b)	M1	For attempting to solve the given 3TQ for the velocity. A correct method must be used, and they must reach **two** values of t for the award of this mark. See General Guidance for the definition of an attempt to solve a 3TQ. If there is no visible method seen, both values of t must be seen for the award of this mark.
	A1	For the two correct values of t They do not need to ne identified as t_1 or t_2 for this mark. Accept $\frac{5}{2}$ and 7 seen.
(c)	M1	For an attempt to integrate the given expression for v. At least one term must be correct and no terms are to be differentiated. If the value of 35 'disappears' it is M0. The question states 'use calculus' so integration must be seen.
	M1	For substituting the **two** values of t into their integrated expression. and subtracting the result (either way) of both substitutions. Allow for any changed expression from v A correct answer of $\pm\frac{243}{8}$ or ± 30.375 implies correct substitution into a correct integral. If the integration is incorrect, or the values are incorrect, full correct substitution must be seen for the award of this mark.
	A1	For the correct distance (which must be a positive value).

Type MS · When is the particle at rest (or at a given velocity)?

Answer 7 of 16 · Question 5 part b · 2 marks

2023 JUN · Paper 02 (regional R) · extract 5 of 5 · find part b in this mark scheme extract

Question	Scheme	Marks

Type MS · When is the particle at rest (or at a given velocity)?

Answer 8 of 16 · Question 5 part a · 3 marks

2024 NOV · Paper 01 · find part a in this mark scheme extract

5 (a)	$3t^2 - 16t + 5 = 0 \rightarrow (3t-1)(t-5) \rightarrow t =$ $t = \frac{1}{3}$ oe $\quad t = 5$	M1 M1 A1 [3]
(b)	$(a =) 6t - 16 > 0$ $t > \frac{8}{3}$ oe	M1 A1 [2]
(c)	$\int_{(1)}^{(4)} (3t^2 - 16t + 5)\, dt$ $\left[t^3 - 8t^2 + 5t\right]_{(1)}^{(4)}$ $\left(4^3 - 8(4)^2 + 5(4)\right) - \left(1^3 - 8(1)^2 + 5(1)\right)$ -42 (distance) = 42	 M1 A1 M1 A1 A1 [5]
		Total 10 marks

Type MS · When is the particle at rest (or at a given velocity)?

Answer 9 of 16 · Question 1 part whole · 4 marks

2020 JAN · Paper 02 · extract 1 of 2 · find part whole in this mark scheme extract

1	$\frac{ds}{dt} = 3t^2 + 8t - 27 = 8$ $3t^2 + 8t - 35\ (= 0)$ $(3t - 7)(t + 5) = 0$ $t = \frac{7}{3}$	M1 A1 M1A1cao **[4]**

M1	Attempt the differentiation and equate their result to 8. Power of at least one term to decrease and none to increase'
A1	Obtain the correct 3TQ. Terms can be in any order and = 0 may be omitted.
M1	Attempt to solve their 3TQ by any valid method. Must reach t = ...
A1cao	For $t = \frac{7}{3}$ (negative answer must be omitted or eliminated) or t = 2.33 or better

Type MS · When is the particle at rest (or at a given velocity)?

Answer 9 of 16 · Question 1 part whole · 4 marks

2020 JAN · Paper 02 · extract 2 of 2 · find part whole in this mark scheme extract

Question Number	Scheme	Marks

Type MS · When is the particle at rest (or at a given velocity)?

Answer 10 of 16 · Question 6 part b · 4 marks

2021 NOV · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

6 (a)	8 (m) (must be positive)	B1 [1]
(b)	$\frac{ds}{dt} = 3t^2 - 8t - 16$ $\frac{ds}{dt} = 0$ when P is stationary $\Rightarrow 3t^2 - 8t - 16 = 0$ $3t^2 - 8t - 16 = (3t + 4)(t - 4) = 0 \Rightarrow t = 4$	M1A1 M1A1 [4]
(c)	$\frac{d^2s}{dt^2} = 6t - 8$ $6t - 8 = 10 \Rightarrow t = 3$	B1ft M1A1 [3]
		Total 8 marks

Type MS · When is the particle at rest (or at a given velocity)?

Answer 10 of 16 · Question 6 part b · 4 marks

2021 NOV · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · When is the particle at rest (or at a given velocity)?

Answer 11 of 16 · Question 6 part c · 3 marks

2021 NOV · Paper 02 · extract 1 of 2 · find part c in this mark scheme extract

6 (a)	8 (m) (must be positive)	B1 [1]
(b)	$\frac{ds}{dt} = 3t^2 - 8t - 16$ $\frac{ds}{dt} = 0$ when P is stationary $\Rightarrow 3t^2 - 8t - 16 = 0$ $3t^2 - 8t - 16 = (3t + 4)(t - 4) = 0 \Rightarrow t = 4$	M1A1 M1A1 [4]
(c)	$\frac{d^2s}{dt^2} = 6t - 8$ $6t - 8 = 10 \Rightarrow t = 3$	B1ft M1A1 [3]
		Total 8 marks

Type MS · When is the particle at rest (or at a given velocity)?

Answer 11 of 16 · Question 6 part c · 3 marks

2021 NOV · Paper 02 · extract 2 of 2 · find part c in this mark scheme extract

Question number	Scheme	Marks

Type MS · When is the particle at rest (or at a given velocity)?

Answer 12 of 16 · Question 2 part b · 2 marks

2022 JUN · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

2 a	$v = \frac{dx}{dt} = 4t^3 - 13.5$	M1
	When $t = 3$ $v = 4(3)^3 - 13.5 = 94.5$ ms^{-1}	A1 (2)
b	$4t^3 - 13.5 = 0 \Rightarrow t^3 = \frac{27}{8} \Rightarrow t = 1.5$	M1 A1ft (2)
c	$a = \frac{dv}{dt} = 12t^2$	M1
	When $t = 2$ $a = 12 \times 2^2 = 48$ ms^{-2}	A1 (2)
		Total 6 marks

Part	Mark	Notes
Ignore incorrect/spurious notation through this question. e.g. ignore $\frac{dy}{dx} = ...$ or the LHS		
(a)	**M1**	For an attempt to differentiate the given expression [with no terms integrated] See General Guidance for the definition of an attempt to differentiate.
	A1	For substituting the value of $t = 3$ into their differentiated expression and obtains 94.5 (units not required)
(b)	**M1**	For setting their $\frac{dx}{dt} = 0$ which must be at a minimum of the form $\pm kt^3 \pm l$ and attempting to find a value for t
	A1ft	For $t = 1.5$ Ft their expression for v which must have come from an acceptable attempt to differentiate the given x
(c)	**M1**	For differentiating their $\frac{dx}{dt}$ which must be of the form $\frac{dx}{dt}$ (or v) $= \pm kt^3 \pm l$ where k and l are constants, [with no term integrated]
	A1	For substituting $t = 2$ into their differentiated expression and obtains 48 (units not required)

Type MS · When is the particle at rest (or at a given velocity)?

Answer 12 of 16 · Question 2 part b · 2 marks

2022 JUN · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Scheme	Marks
	$\left\|\overrightarrow{OA}\right\| = \sqrt{\quad}$ $\left\|\overrightarrow{OB}\right\| = \sqrt{\quad}$	

Type MS · When is the particle at rest (or at a given velocity)?

Answer 13 of 16 · Question 3 part a · 3 marks

2025 JUN · Paper 01 (regional R) · extract 1 of 3 · find part a in this mark scheme extract

ALT for first 3 marks: Uses the third term		
	B1	Let the 3rd term be T $20-T=d \quad T-8=d$
	M1	Solves the above equations simultaneously by any method to find d **AND** finds a value for a
	A1	For both correct $a=-4$ and $d=6$. Values may be implied by subsequent working
	M1	Correct substitution of their values for a and d into the correct formula for S_n (doesn't need to be placed > 220 at this point)
	M1	For the correct inequality shown with their values oe
	M1	For a full and correct method to solve their 3TQ leading to at least one value for n. Award this mark for the positive root [9.809…] without explicit working seen [use of calculator] following a correct 3TQ seen. If the 3TQ is incorrect and there is no evidence of explicit working to solve their 3TQ do not award this mark.
	A1	For choosing only positive root, and rounding to $n = 10$ If a candidate offers a positive and negative value without rejecting the negative [-7.47…] do not award this mark.
ALT: uses T & I Final two marks		
	M1	Attempts $n = 10$ **AND** $n = 9$ When $n = 10$ $S = 230$ and when $n = 9$ $S = 180$ **These values must be seen**
	A1	For choosing $n = 10$

Type MS · When is the particle at rest (or at a given velocity)?

Answer 13 of 16 · Question 3 part a · 3 marks

2025 JUN · Paper 01 (regional R) · extract 2 of 3 · find part a in this mark scheme extract

Question number	Scheme	Marks
3 (a)	$\left(v = 2t^2 - 19t + 35\right)$	
	$\left(\frac{dv}{dt} =\right) 4t - 19$	M1
	$4t - 19 = 0$	M1
	$t = \frac{19}{4}$	A1 [3]
(b)	$\left(v = 0 = 2t^2 - 19t + 35\right)$	
	$0 = (2t - 5)(t - 7)$	M1
	$T_1 = \frac{5}{2}$, $T_2 = 7$	A1, A1 [3]
(c)	$\left(\int\left(2t^2 - 19t + 35\right)dt\right) = \frac{2}{3}t^3 - \frac{19}{2}t^2 + 35t(+C)$ oe	M1 A1
	OR	
	$\int_{'2.5'}^{'7'}\left(2t^2 - 19t + 35\right)\,dt = \left[\frac{2}{3}t^3 - \frac{19}{2}t^2 + 35t\right]_{'2.5'}^{'7'}$	
	$\left(\frac{2}{3}(7)^3 - \frac{19}{2}(7)^2 + 35(7)\right) - \left(\frac{2}{3}\left(\frac{5}{2}\right)^3 - \frac{19}{2}\left(\frac{5}{2}\right)^2 + 35\left(\frac{5}{2}\right)\right) = -\frac{243}{8}$	M1
	$d = \frac{243}{8}$	A1 [4]
		Total 10 marks

Type MS · When is the particle at rest (or at a given velocity)?

Answer 13 of 16 · Question 3 part a · 3 marks

2025 JUN · Paper 01 (regional R) · extract 3 of 3 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · When is the particle at rest (or at a given velocity)?

Answer 14 of 16 · Question 3 part b · 3 marks

2025 JUN · Paper 01 (regional R) · extract 1 of 3 · find part b in this mark scheme extract

ALT for first 3 marks: Uses the third term		
	B1	Let the 3rd term be T $20-T=d \quad T-8=d$
	M1	Solves the above equations simultaneously by any method to find d **AND** finds a value for a
	A1	For both correct $a=-4$ and $d=6$. Values may be implied by subsequent working
	M1	Correct substitution of their values for a and d into the correct formula for S_n (doesn't need to be placed > 220 at this point)
	M1	For the correct inequality shown with their values oe
	M1	For a full and correct method to solve their 3TQ leading to at least one value for n. Award this mark for the positive root [9.809…] without explicit working seen [use of calculator] following a correct 3TQ seen. If the 3TQ is incorrect and there is no evidence of explicit working to solve their 3TQ do not award this mark.
	A1	For choosing only positive root, and rounding to $n = 10$ If a candidate offers a positive and negative value without rejecting the negative [-7.47…] do not award this mark.
ALT: uses T & I Final two marks		
	M1	Attempts $n = 10$ **AND** $n = 9$ When $n = 10$ $S = 230$ and when $n = 9$ $S = 180$ **These values must be seen**
	A1	For choosing $n = 10$

Type MS · When is the particle at rest (or at a given velocity)?

Answer 14 of 16 · Question 3 part b · 3 marks

2025 JUN · Paper 01 (regional R) · extract 2 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks
3 (a)	$\left(v = 2t^2 - 19t + 35\right)$	
	$\left(\frac{dv}{dt} =\right) 4t - 19$	M1
	$4t - 19 = 0$	M1
	$t = \frac{19}{4}$	A1 [3]
(b)	$\left(v = 0 = 2t^2 - 19t + 35\right)$	
	$0 = (2t - 5)(t - 7)$	M1
	$T_1 = \frac{5}{2}$, $T_2 = 7$	A1, A1 [3]
(c)	$\left(\int\left(2t^2 - 19t + 35\right)dt\right) = \frac{2}{3}t^3 - \frac{19}{2}t^2 + 35t(+C)$ oe	M1 A1
	OR	
	$\int_{'2.5'}^{'7'}\left(2t^2 - 19t + 35\right)\,dt = \left[\frac{2}{3}t^3 - \frac{19}{2}t^2 + 35t\right]_{'2.5'}^{'7'}$	
	$\left(\frac{2}{3}(7)^3 - \frac{19}{2}(7)^2 + 35(7)\right) - \left(\frac{2}{3}\left(\frac{5}{2}\right)^3 - \frac{19}{2}\left(\frac{5}{2}\right)^2 + 35\left(\frac{5}{2}\right)\right) = -\frac{243}{8}$	M1
	$d = \frac{243}{8}$	A1 [4]
		Total 10 marks

Type MS · When is the particle at rest (or at a given velocity)?

Answer 14 of 16 · Question 3 part b · 3 marks

2025 JUN · Paper 01 (regional R) · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · When is the particle at rest (or at a given velocity)?

Answer 15 of 16 · Question 5 part b · 2 marks

2022 JAN · Paper 02 · extract 1 of 3 · find part b in this mark scheme extract

5(a)	$3t^2-23t+30=0\Rightarrow(3t-5)(t-6)=0\Rightarrow t=6,\quad \frac{5}{3}$	M1A1A1 [3]
(b)	$a=\frac{\mathrm{d}v}{\mathrm{d}t}=6t-23$ $(6t-23>0)\Rightarrow t>\frac{23}{6}$	M1 A1 [2]
(c)	$s=\int 3t^2-23t+30\ \mathrm{d}t=\frac{3t^3}{3}-\frac{23t^2}{2}+30t+c$ $26=\frac{3(8)^3}{3}-\frac{23(8)^2}{2}+30(8)+c\Rightarrow c='10'$ $d=0+0+0+'10'\Rightarrow d=10$	M1A1 M1 A1 [4]
	Total	**9 marks**

Type MS · When is the particle at rest (or at a given velocity)?

Answer 15 of 16 · Question 5 part b · 2 marks

2022 JAN · Paper 02 · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Notes
5 (a)	M1	For setting the given expression for $v = 0$ and attempting to solve the 3TQ See general guidance for the definition of an attempt $3t^2 - 23t + 30 = 0 \Rightarrow (3t-5)(t-6) = 0 \Rightarrow t = ..., ...$
	A1	For either $t = 6$ or $t = \frac{5}{3}$
	A1	For both $t = 6$ and $t = \frac{5}{3}$
(b)	M1	For differentiating the given expression for v which must be correct for this mark. $a = \frac{dv}{dt} = 6t - 23$
	A1	$(6t - 23 > 0) \Rightarrow t > \frac{23}{6}$ Accept equivalent exact values including $t > 3.8\dot{3}$
(c)	M1	For attempting to integrate the given expression for v See General Guidance for the definition of an attempt with no terms differentiated. $+c$ is not required for the award of this mark. $s = \int 3t^2 - 23t + 30 \, dt = \frac{3t^3}{3} - \frac{23t^2}{2} + 30t + c$
	A1	For the correct integrated expression, which must include a constant term e.g.+ c $s = \frac{3t^3}{3} - \frac{23t^2}{2} + 30t + c$ Simplification is not required for this mark
	M1	For substituting the given values of $t = 8$ when $s = 26$ into their **integrated** expression to find c $26 = \frac{3(8)^3}{3} - \frac{23(8)^2}{2} + 30(8) + c \Rightarrow c = '10'$ $[d = 0 + 0 + 0 + '10' \Rightarrow d = ...]$
	A1	For the correct value of $d = 10$

Type MS · When is the particle at rest (or at a given velocity)?

Answer 15 of 16 · Question 5 part b · 2 marks

2022 JAN · Paper 02 · extract 3 of 3 · find part b in this mark scheme extract

Question	Scheme	Mark

Type MS · When is the particle at rest (or at a given velocity)?

Answer 16 of 16 · Question 5 part b · 2 marks

2024 NOV · Paper 01 · find part b in this mark scheme extract

5 (a)	$3t^2 - 16t + 5 = 0 \rightarrow (3t-1)(t-5) \rightarrow t =$ $t = \frac{1}{3}$ oe $\quad t = 5$	M1 M1 A1 [3]
(b)	$(a =) 6t - 16 > 0$ $t > \frac{8}{3}$ oe	M1 A1 [2]
(c)	$\int_{(1)}^{(4)} (3t^2 - 16t + 5)\, dt$ $\left[t^3 - 8t^2 + 5t\right]_{(1)}^{(4)}$ $\left(4^3 - 8(4)^2 + 5(4)\right) - \left(1^3 - 8(1)^2 + 5(1)\right)$ -42 (distance) = 42	 M1 A1 M1 A1 A1 [5]
		Total 10 marks